

Question ID 29c177e6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 29c177e6

SAT3 M 2.1

What is 10% of 470?

- A. 37
- B. 47
- C. 423
- D. 460

ID: 29c177e6 Answer

Correct Answer: B

Rationale

Choice B is correct. 10% of a quantity means $\frac{10}{100}$ times the quantity. Therefore, 10% of 470 can be represented as $\frac{10}{100}(470)$, which is equivalent to $0.10(470)$, or 47. Therefore, 10% of 470 is 47.

Choice A is incorrect. This is 10% of 370, not 10% of 470.

Choice C is incorrect. This is 90% of 470, not 10% of 470.

Choice D is incorrect. This is $470 - 10$, not 10% of 470.

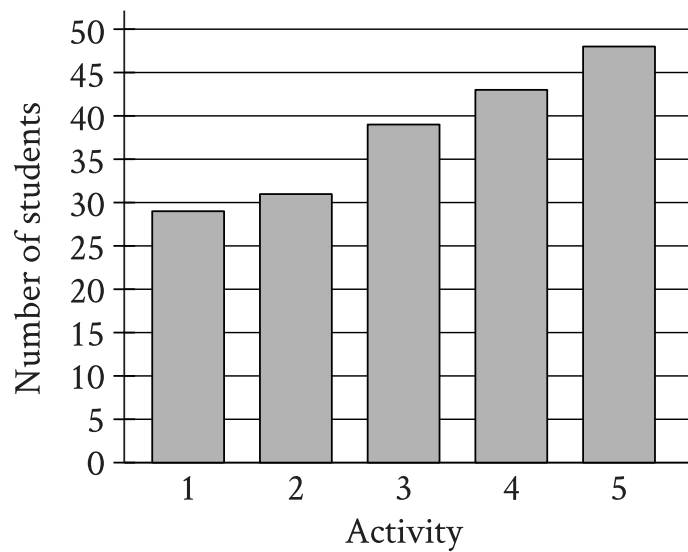
Question Difficulty: Easy

Question ID 93779b53

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 93779b53

SAT3 M 2.2



A group of students voted on five after-school activities. The bar graph shows the number of students who voted for each of the five activities. How many students chose activity **3**?

- A. **25**
- B. **39**
- C. **48**
- D. **50**

ID: 93779b53 Answer

Correct Answer: B

Rationale

Choice B is correct. The height of each bar in the bar graph given represents the number of students that voted for the activity specified at the bottom of the bar. The bar for activity **3** has a height that is between **35** and **40**. In other words, the number of students that chose activity **3** is between **35** students and **40** students. Of the given choices, **39** is the only value between **35** and **40**. Therefore, **39** students chose activity **3**.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the number of students that chose activity **5**, not activity **3**.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 349a5bc1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 349a5bc1

SAT3 M 2.3

$4x + 5 = 165$

What is the solution to the given equation?

ID: 349a5bc1 Answer

Correct Answer: 40

Rationale

The correct answer is 40. Subtracting 5 from both sides of the given equation yields $4x = 160$. Dividing both sides of this equation by 4 yields $x = 40$. Therefore, the solution to the given equation is 40.

Question Difficulty: Easy

Question ID 86636d8f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 86636d8f

SAT3 M 2.4

A customer spent **\$27** to purchase oranges at **\$3** per pound. How many pounds of oranges did the customer purchase?

ID: 86636d8f Answer

Correct Answer: 9

Rationale

The correct answer is **9**. It’s given that the customer spent **\$27** to purchase oranges at **\$3** per pound. Therefore, the number of pounds of oranges the customer purchased is $\$27\left(\frac{1\text{ pound}}{\$3}\right)$, or **9** pounds.

Question Difficulty: Easy

Question ID 0d391910

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0d391910

SAT3 M 2.5

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

ID: 0d391910 Answer

Correct Answer: 2

Rationale

The correct answer is 2. Substituting 8 for $f(x)$ in the given equation yields $8 = 4x$. Dividing the left- and right-hand sides of this equation by 4 yields $x = 2$. Therefore, the value of x is 2 when $f(x) = 8$.

Question Difficulty: Easy

Question ID 3ea87153

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 3ea87153

SAT3 M 2.6

The function g is defined by $g(x) = x^2 + 9$. For which value of x is $g(x) = 25$?

- A. 4
- B. 5
- C. 9
- D. 13

ID: 3ea87153 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $g(x) = x^2 + 9$. Substituting 25 for $g(x)$ in this equation yields $25 = x^2 + 9$. Subtracting 9 from both sides of this equation yields $16 = x^2$. Taking the square root of each side of this equation yields $x = \pm 4$. It follows that $g(x) = 25$ when the value of x is 4 or -4 . Only 4 is listed among the choices.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

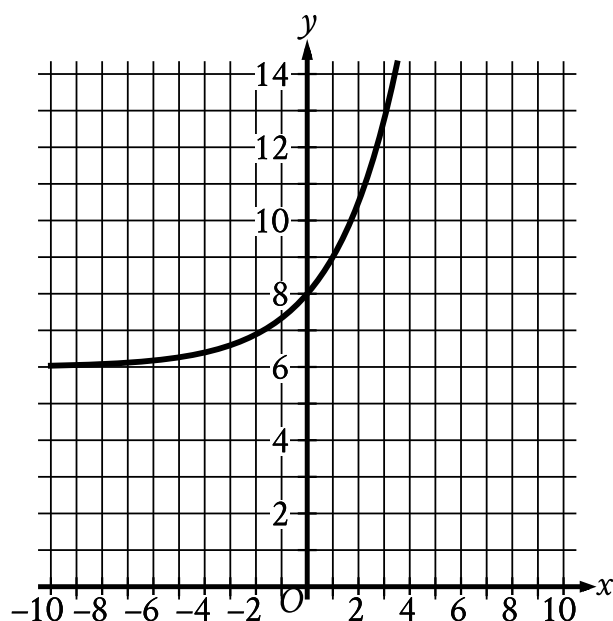
Question Difficulty: Easy

Question ID f547a8b1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f547a8b1

SAT3 M 2.7



What is the y -intercept of the graph shown?

- A. $(-8, 0)$
- B. $(-6, 0)$
- C. $(0, 6)$
- D. $(0, 8)$

ID: f547a8b1 Answer

Correct Answer: D

Rationale

Choice D is correct. The y -intercept of a graph in the xy -plane is the point at which the graph crosses the y -axis. The graph shown crosses the y -axis at the point $(0, 8)$. Therefore, the y -intercept of the graph shown is $(0, 8)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID b8cbe394

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: b8cbe394

SAT3 M 2.8

Sean rents a tent at a cost of **\$11** per day plus a onetime insurance fee of **\$10**. Which equation represents the total cost c , in dollars, to rent the tent with insurance for d days?

- A. $c = 11(d + 10)$
- B. $c = 10(d + 11)$
- C. $c = 11d + 10$
- D. $c = 10d + 11$

ID: b8cbe394 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the cost of renting a tent is **\$11** per day for d days. Multiplying the rental cost by the number of days yields **\$11 d** , which represents the cost of renting the tent for d days before the insurance is added. Adding the onetime insurance fee of **\$10** to the rental cost of **\$11 d** gives the total cost c , in dollars, which can be represented by the equation $c = 11d + 10$.

Choice A is incorrect. This equation represents the total cost to rent the tent if the insurance fee was charged every day.

Choice B is incorrect. This equation represents the total cost to rent the tent if the daily fee was **\$($d + 11$)** for **10** days.

Choice D is incorrect. This equation represents the total cost to rent the tent if the daily fee was **\$10** and the onetime fee was **\$11**.

Question Difficulty: Easy

Question ID 433184f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	■ ■ ■

ID: 433184f1

SAT3 M 2.9

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

- A. $\frac{1}{(x+1)(4x-5)}$
- B. $\frac{3}{3x-6}$
- C. $-\frac{1}{(x+1)(4x-5)}$
- D. $\frac{9}{(x+1)(4x-5)}$

ID: 433184f1 Answer

Correct Answer: D

Rationale

Choice D is correct. The expression $\frac{4}{4x-5} - \frac{1}{x+1}$ can be rewritten as $\frac{4}{4x-5} + \frac{(-1)}{x+1}$. To add the two terms of this expression, the terms can be rewritten with a common denominator. Since $\frac{x+1}{x+1} = 1$, the expression $\frac{4}{4x-5}$ can be rewritten as $\frac{(x+1)(4)}{(x+1)(4x-5)}$. Since $\frac{4x-5}{4x-5} = 1$, the expression $\frac{-1}{x+1}$ can be rewritten as $\frac{(4x-5)(-1)}{(4x-5)(x+1)}$. Therefore, the expression $\frac{4}{4x-5} + \frac{(-1)}{x+1}$ can be rewritten as $\frac{(x+1)(4)}{(x+1)(4x-5)} + \frac{(4x-5)(-1)}{(4x-5)(x+1)}$, which is equivalent to $\frac{(x+1)(4) + (4x-5)(-1)}{(x+1)(4x-5)}$. Applying the distributive property to each term of the numerator yields $\frac{(4x+4) + (-4x+5)}{(x+1)(4x-5)}$, or $\frac{(4x+(-4x)) + (4+5)}{(x+1)(4x-5)}$. Adding like terms in the numerator yields $\frac{9}{(x+1)(4x-5)}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

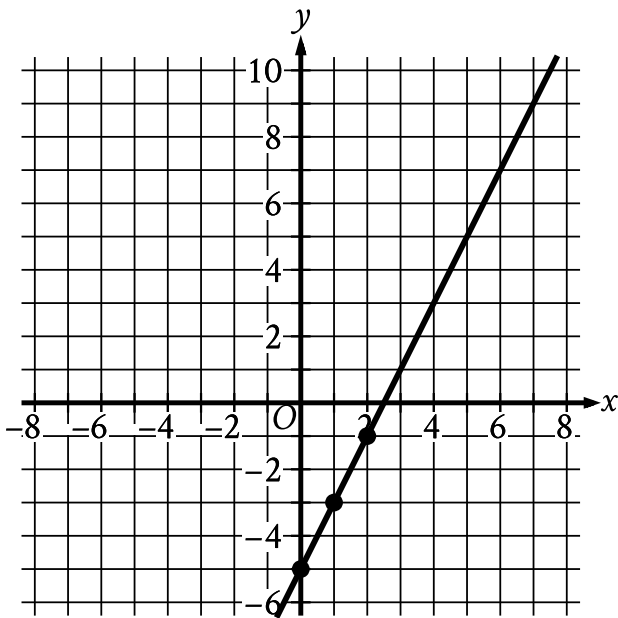
Question Difficulty: Hard

Question ID 4acd05cd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: 4acd05cd

SAT3 M 2.10



The graph shows the linear relationship between x and y . Which table gives three values of x and their corresponding values of y for this relationship?

A.

x	y
0	0
1	-7
2	-9

B.

x	y
0	0
1	-3
2	-1

C.

x	y
0	-5
1	-7

2	−9
----------	-----------

D.

x	y
0	−5
1	−3
2	−1

ID: 4acd05cd Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the graph shows the linear relationship between x and y . The given graph passes through the points $(0, -5)$, $(1, -3)$, and $(2, -1)$. It follows that when $x = 0$, the corresponding value of y is -5 , when $x = 1$, the corresponding value of y is -3 , and when $x = 2$, the corresponding value of y is -1 . Of the given choices, only the table in choice D gives these three values of x and their corresponding values of y for the relationship shown in the graph.

Choice A is incorrect. This table represents a relationship between x and y such that the graph passes through the points $(0, 0)$, $(1, -7)$, and $(2, -9)$.

Choice B is incorrect. This table represents a relationship between x and y such that the graph passes through the points $(0, 0)$, $(1, -3)$, and $(2, -1)$.

Choice C is incorrect. This table represents a linear relationship between x and y such that the graph passes through the points $(0, -5)$, $(1, -7)$, and $(2, -9)$.

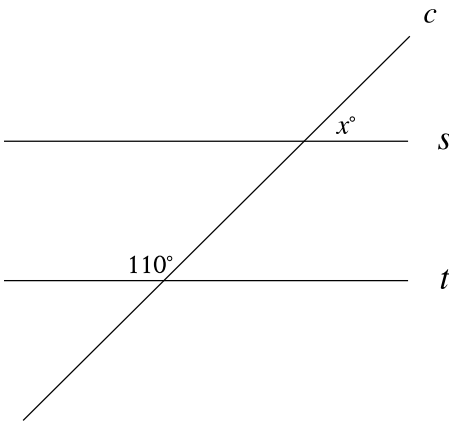
Question Difficulty: Easy

Question ID cf0d3050

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: cf0d3050

SAT3 M 2.11



Note: Figure not drawn to scale.

In the figure shown, line *c* intersects parallel lines *s* and *t*. What is the value of *x*?

ID: cf0d3050 Answer

Correct Answer: 70

Rationale

The correct answer is **70**. Based on the figure, the angle with measure **110°** and the angle vertical to the angle with measure *x*° are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure *x*° also has measure *x*°. It's given that lines *s* and *t* are parallel. Therefore, same side interior angles between lines *s* and *t* are supplementary. It follows that *x* + 110 = 180. Subtracting 110 from both sides of this equation yields *x* = 70.

Question Difficulty: Easy

Question ID 02b02213

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 02b02213

SAT3 M 2.12

What is the perimeter, in inches, of a rectangle with a length of **4** inches and a width of **9** inches?

- A. **13**
- B. **17**
- C. **22**
- D. **26**

ID: 02b02213 Answer

Correct Answer: D

Rationale

Choice D is correct. The perimeter of a figure is equal to the sum of the measurements of the sides of the figure. It's given that the rectangle has a length of **4** inches and a width of **9** inches. Since a rectangle has **4** sides, of which opposite sides are parallel and equal, it follows that the rectangle has two sides with a length of **4** inches and two sides with a width of **9** inches. Therefore, the perimeter of this rectangle is $4 + 4 + 9 + 9$, or **26** inches.

Choice A is incorrect. This is the sum, in inches, of the length and the width of the rectangle.

Choice B is incorrect. This is the sum, in inches, of the two lengths and the width of the rectangle.

Choice C is incorrect. This is the sum, in inches, of the length and the two widths of the rectangle.

Question Difficulty: Easy

Question ID b8c4a1cd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b8c4a1cd

SAT3 M 2.13

$8j = k + 15m$

The given equation relates the distinct positive numbers j , k , and m . Which equation correctly expresses j in terms of k and m ?

- A. $j = \frac{k}{8} + 15m$
- B. $j = k + \frac{15m}{8}$
- C. $j = 8(k + 15m)$
- D. $j = \frac{k+15m}{8}$

ID: b8c4a1cd Answer

Correct Answer: D

Rationale

Choice D is correct. To express j in terms of k and m , the given equation must be solved for j . Dividing each side of the given equation by 8 yields $j = \frac{k+15m}{8}$.

Choice A is incorrect. This is equivalent to $8j = k + 120m$.

Choice B is incorrect. This is equivalent to $8j = 8k + 15m$.

Choice C is incorrect. This is equivalent to $\frac{j}{8} = k + 15m$.

Question Difficulty: Easy

Question ID cfe67646

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: cfe67646

SAT3 M 2.14

The point $(8, 2)$ in the xy -plane is a solution to which of the following systems of inequalities?

- A. $x > 0$
 $y > 0$
- B. $x > 0$
 $y < 0$
- C. $x < 0$
 $y > 0$
- D. $x < 0$
 $y < 0$

ID: cfe67646 Answer

Correct Answer: A

Rationale

Choice A is correct. The given point, $(8, 2)$, is located in the first quadrant in the xy -plane. The system of inequalities in choice A represents all the points in the first quadrant in the xy -plane. Therefore, $(8, 2)$ is a solution to the system of inequalities in choice A.

Alternate approach: Substituting 8 for x in the first inequality in choice A, $x > 0$, yields $8 > 0$, which is true. Substituting 2 for y in the second inequality in choice A, $y > 0$, yields $2 > 0$, which is true. Since the coordinates of the point $(8, 2)$ make the inequalities $x > 0$ and $y > 0$ true, the point $(8, 2)$ is a solution to the system of inequalities consisting of $x > 0$ and $y > 0$.

Choice B is incorrect. This system of inequalities represents all the points in the fourth quadrant, not the first quadrant, in the xy -plane.

Choice C is incorrect. This system of inequalities represents all the points in the second quadrant, not the first quadrant, in the xy -plane.

Choice D is incorrect. This system of inequalities represents all the points in the third quadrant, not the first quadrant, in the xy -plane.

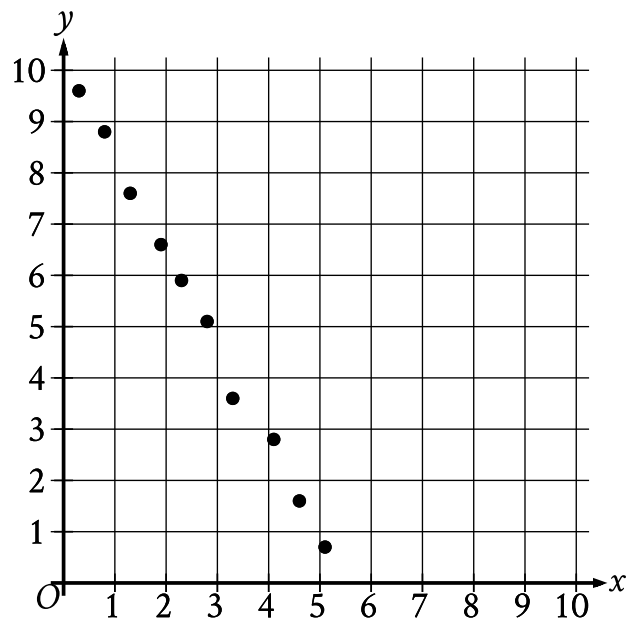
Question Difficulty: Easy

Question ID 5f3ee607

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 5f3ee607

SAT3 M 2.15



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

- A. $y = -1.9x - 10.1$
- B. $y = -1.9x + 10.1$
- C. $y = 1.9x - 10.1$
- D. $y = 1.9x + 10.1$

ID: 5f3ee607 Answer

Correct Answer: B

Rationale

Choice B is correct. The equation representing a linear model can be written in the form $y = a + bx$, or $y = bx + a$, where b is the slope of the graph of the model and $(0, a)$ is the y -intercept of the graph of the model. The scatterplot shows that as the x -values of the data points increase, the y -values of the data points decrease, which means the graph of an appropriate linear model has a negative slope. Therefore, $b < 0$. The scatterplot also shows that the data points are close to the y -axis at a positive value of y . Therefore, the y -intercept of the graph of an appropriate linear model has a positive y -coordinate, which means $a > 0$. Of the given choices, only choice B, $y = -1.9x + 10.1$, has a negative value for b , the slope, and a positive value for a , the y -coordinate of the y -intercept.

Choice A is incorrect. The graph of this model has a y -intercept with a negative y -coordinate, not a positive y -coordinate.

Choice C is incorrect. The graph of this model has a positive slope, not a negative slope, and a y -intercept with a negative y -coordinate, not a positive y -coordinate.

Choice D is incorrect. The graph of this model has a positive slope, not a negative slope.

Question Difficulty: Easy

Question ID d8ace155

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: d8ace155

SAT3 M 2.16

A company opens an account with an initial balance of **\$36,100.00**. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function **A** , where **$A(t)$** is the account balance, in dollars, **t** years after the account is opened. The account balance after **13** years is **\$68,071.93**. Which equation could define **A** ?

- A. $A(t) = 36,100.00(1.05)^t$
- B. $A(t) = 31,971.93(1.05)^t$
- C. $A(t) = 31,971.93(0.05)^t$
- D. $A(t) = 36,100.00(0.05)^t$

ID: d8ace155 Answer

Correct Answer: A

Rationale

Choice A is correct. Since it's given that the account balance, **$A(t)$** , in dollars, after **t** years can be modeled by an exponential function, it follows that function **A** can be written in the form **$A(t) = Nr^t$** , where **N** is the initial value of the function and **r** is a constant related to the growth of the function. It's given that the initial balance of the account is **\$36,100.00**, so it follows that the initial value of the function, or **N** , must be **36,100.00**. Substituting **36,100.00** for **N** in the equation **$A(t) = Nr^t$** yields **$A(t) = 36,100.00r^t$** . It's given that the account balance after **13** years, or when **$t = 13$** , is **\$68,071.93**. It follows that **$A(13) = 68,071.93$** , or **$36,100.00r^{13} = 68,071.93$** . Dividing each side of the equation **$36,100.00r^{13} = 68,071.93$** by **36,100.00** yields **$r^{13} = \frac{68,071.93}{36,100.00}$** . Taking the **13th** root of both sides of this equation yields **$r = \sqrt[13]{\frac{68,071.93}{36,100.00}}$** , or **$r$** is approximately equal to **1.05**. Substituting **1.05** for **r** in the equation **$A(t) = 36,100.00r^t$** yields **$A(t) = 36,100.00(1.05)^t$** , so the equation **$A(t) = 36,100.00(1.05)^t$** could define **A** .

Choice B is incorrect. Substituting **0** for **t** in this function indicates an initial balance of **\$31,971.93**, rather than **\$36,100.00**.

Choice C is incorrect. Substituting **0** for **t** in this function indicates an initial balance of **\$31,971.93**, rather than **\$36,100.00**. Additionally, this function indicates the account balance is decreasing, rather than increasing, over time.

Choice D is incorrect. This function indicates the account balance is decreasing, rather than increasing, over time.

Question Difficulty: Medium

Question ID 3a9d60b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: 3a9d60b2

SAT3 M 2.17

$$2|4 - x| + 3|4 - x| = 25$$

What is the positive solution to the given equation?

ID: 3a9d60b2 Answer

Correct Answer: 9

Rationale

The correct answer is 9. The given equation can be rewritten as $5|4 - x| = 25$. Dividing each side of this equation by 5 yields $|4 - x| = 5$. By the definition of absolute value, if $|4 - x| = 5$, then $4 - x = 5$ or $4 - x = -5$. Subtracting 4 from each side of the equation $4 - x = 5$ yields $-x = 1$. Dividing each side of this equation by -1 yields $x = -1$. Similarly, subtracting 4 from each side of the equation $4 - x = -5$ yields $-x = -9$. Dividing each side of this equation by -1 yields $x = 9$. Therefore, since the two solutions to the given equation are -1 and 9 , the positive solution to the given equation is 9 .

Question Difficulty: Hard

Question ID b4a6ed81

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: b4a6ed81

SAT3 M 2.18

The expression $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, where r is a constant. What is the value of r ?

ID: b4a6ed81 Answer

Correct Answer: 6

Rationale

The correct answer is 6. Applying the distributive property to the expression $ry^4(15y - 9)$ yields $15ry^5 - 9ry^4$. Since $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, it follows that $90y^5 - 54y^4$ is also equivalent to $15ry^5 - 9ry^4$. Since these expressions are equivalent, it follows that corresponding coefficients are equivalent. Therefore, $90 = 15r$ and $-54 = -9r$. Solving either of these equations for r will yield the value of r . Dividing both sides of $90 = 15r$ by 15 yields $6 = r$. Therefore, the value of r is 6.

Question Difficulty: Medium

Question ID 4993b828

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 4993b828

SAT3 M 2.19

The area A , in square centimeters, of a rectangular cutting board can be represented by the expression $w(w + 9)$, where w is the width, in centimeters, of the cutting board. Which expression represents the length, in centimeters, of the cutting board?

- A. $w(w + 9)$
- B. w
- C. 9
- D. $(w + 9)$

ID: 4993b828 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the expression $w(w + 9)$ represents the area, in square centimeters, of a rectangular cutting board, where w is the width, in centimeters, of the cutting board. The area of a rectangle can be calculated by multiplying its length by its width. It follows that the length, in centimeters, of the cutting board is represented by the expression $(w + 9)$.

Choice A is incorrect. This expression represents the area, in square centimeters, of the cutting board, not its length, in centimeters.

Choice B is incorrect. This expression represents the width, in centimeters, of the cutting board, not its length.

Choice C is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the cutting board, not its length, in centimeters.

Question Difficulty: Medium

Question ID 48fb34c8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	■ ■ ■

ID: 48fb34c8

SAT3 M 2.20

$$y > 13x - 18$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	21
5	47
8	86

B.

x	y
3	26
5	42
8	86

C.

x	y
3	16
5	42
8	81

D.

x	y
3	26
5	52
8	91

ID: 48fb34c8 Answer

Correct Answer: D

Rationale

Choice D is correct. All the tables in the choices have the same three values of x , so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables.

Substituting **3** for x in the given inequality yields $y > 13(3) - 18$, or $y > 21$. Therefore, when $x = 3$, the corresponding value of y is greater than **21**. Substituting **5** for x in the given inequality yields $y > 13(5) - 18$, or $y > 47$. Therefore, when $x = 5$, the corresponding value of y is greater than **47**. Substituting **8** for x in the given inequality yields $y > 13(8) - 18$, or $y > 86$. Therefore, when $x = 8$, the corresponding value of y is greater than **86**. For the table in choice D, when $x = 3$, the corresponding value of y is **26**, which is greater than **21**; when $x = 5$, the corresponding value of y is **52**, which is greater than **47**; when $x = 8$, the corresponding value of y is **91**, which is greater than **86**. Therefore, the table in choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. In the table for choice A, when $x = 3$, the corresponding value of y is **21**, which is not greater than **21**; when $x = 5$, the corresponding value of y is **47**, which is not greater than **47**; when $x = 8$, the corresponding value of y is **86**, which is not greater than **86**.

Choice B is incorrect. In the table for choice B, when $x = 5$, the corresponding value of y is **42**, which is not greater than **47**; when $x = 8$, the corresponding value of y is **86**, which is not greater than **86**.

Choice C is incorrect. In the table for choice C, when $x = 3$, the corresponding value of y is **16**, which is not greater than **21**; when $x = 5$, the corresponding value of y is **42**, which is not greater than **47**; when $x = 8$, the corresponding value of y is **81**, which is not greater than **86**.

Question Difficulty: Hard

Question ID cd358b89

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■ ■ ■

ID: cd358b89

SAT3 M 2.21

Function f is defined by $f(x) = (x + 6)(x + 5)(x + 1)$. Function g is defined by $g(x) = f(x - 1)$. The graph of $y = g(x)$ in the xy -plane has x -intercepts at $(a, 0)$, $(b, 0)$, and $(c, 0)$, where a , b , and c are distinct constants. What is the value of $a + b + c$?

- A. -15
- B. -9
- C. 11
- D. 15

ID: cd358b89 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $g(x) = f(x - 1)$. Since $f(x) = (x + 6)(x + 5)(x + 1)$, it follows that $f(x - 1) = (x - 1 + 6)(x - 1 + 5)(x - 1 + 1)$. Combining like terms yields $f(x - 1) = (x + 5)(x + 4)(x)$. Therefore, $g(x) = x(x + 5)(x + 4)$. The x -intercepts of a graph in the xy -plane are the points where $y = 0$. The x -coordinates of the x -intercepts of the graph of $y = g(x)$ in the xy -plane can be found by solving the equation $0 = x(x + 5)(x + 4)$. Applying the zero product property to this equation yields three equations: $x = 0$, $x + 5 = 0$, and $x + 4 = 0$. Solving each of these equations for x yields $x = 0$, $x = -5$, and $x = -4$, respectively. Therefore, the x -intercepts of the graph of $y = g(x)$ are $(0, 0)$, $(-5, 0)$, and $(-4, 0)$. It follows that the values of a , b , and c are 0 , -5 , and -4 . Thus, the value of $a + b + c$ is $0 + (-5) + (-4)$, which is equal to -9 .

Choice A is incorrect. This is the value of $a + b + c$ if $g(x) = f(x + 1)$.

Choice C is incorrect. This is the value of $a + b + c - 1$ if $g(x) = (x - 6)(x - 5)(x - 1)$.

Choice D is incorrect. This is the value of $a + b + c$ if $f(x) = (x - 6)(x - 5)(x - 1)$.

Question Difficulty: Hard

Question ID ae041e52

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■ ■ ■

ID: ae041e52

SAT3 M 2.22

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

- A. 20
- B. $\frac{20\sqrt{2}}{2}$
- C. $20\sqrt{2}$
- D. 40

ID: ae041e52 Answer

Correct Answer: A

Rationale

Choice A is correct. When a square is inscribed in a circle, a diagonal of the square is a diameter of the circle. It's given that a square is inscribed in a circle and the length of a radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. Therefore, the length of a diameter of the circle is $2\left(\frac{20\sqrt{2}}{2}\right)$ inches, or $20\sqrt{2}$ inches. It follows that the length of a diagonal of the square is $20\sqrt{2}$ inches. A diagonal of a square separates the square into two right triangles in which the legs are the sides of the square and the hypotenuse is a diagonal. Since a square has 4 congruent sides, each of these two right triangles has congruent legs and a hypotenuse of length $20\sqrt{2}$ inches. Since each of these two right triangles has congruent legs, they are both 45-45-90 triangles. In a 45-45-90 triangle, the length of the hypotenuse is $\sqrt{2}$ times the length of a leg. Let s represent the length of a leg of one of these 45-45-90 triangles. It follows that $20\sqrt{2} = \sqrt{2}(s)$. Dividing both sides of this equation by $\sqrt{2}$ yields $20 = s$. Therefore, the length of a leg of one of these 45-45-90 triangles is 20 inches. Since the legs of these two 45-45-90 triangles are the sides of the square, it follows that the side length of the square is 20 inches.

Choice B is incorrect. This is the length of a radius, in inches, of the circle.

Choice C is incorrect. This is the length of a diameter, in inches, of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard